Expectated Values

1. CB 2.14
2. CB 2.17 *(for part (b) remember that* $\int \frac{1}{1+x^2}dx = \tan^{-1}(x) + constant$*).*
3. CB 2.20
4. CB 2.24
5. Assume $X \sim \text{Gamma}(1/2, \beta)$.
   (a) What is $\mathbb{E}\left(X^{1/2}\right)$? How does $\mathbb{E}\left(X^{1/2}\right)$ compare to $\mathbb{E}(X)^{1/2}$?
   (b) Without solving for the expected value, how do $\mathbb{E}(X^2)$ and $\mathbb{E}(X)^2$ compare?
   (c) Without solving for the expected value, how do $\mathbb{E}(\log X)$ and $\log(\mathbb{E}(X))$ compare?
6. Assume $X \sim \text{Normal}(\mu, \sigma^2)$. What is $\mathbb{E}\left(e^X\right)$? How does $\mathbb{E}\left(e^X\right)$ compare to $e^{\mathbb{E}(X)}$?
7. Let X be a continuous random variable with the density function

   $$f(x) = 2x, \text{ for } 0 \leq x \leq 1$$

   (a) Find $E[X]$.
   (b) Find $E[X^2]$ and $Var(X)$.
   (c) Find $E[4X + 9]$.
   (d) Find $E[X(1 - X)]$.
8. Let $X \sim Beta(\alpha, \beta)$, where $0 \leq x \leq 1$ with $\alpha > 0$ and $\beta > 0$. Find $E[X]$ and $Var(X)$.

Moment Generating Functions

9. CB 2.30*
10. CB 2.33
11. CB 2.38*
12. Let $Z \sim N(0, 1)$. Then the mgf of Z is $M_Z(t) = e^{t^2/2}$. If $X = \mu + \sigma Z$ (where $-\infty \leq \mu \leq \infty$ and $\sigma > 0$ are constants) , then $X \sim N(\mu, \sigma^2)$.
    (a) Find the mgf of X, $M_X(t)$.
    (b) Find the mgf of X/n where $n > 0$ is a constant.
13. Consider random variable, X, with pdf

    $$f(x) = \begin{cases} 1-p & \text{for } x = 0 \\ pe^{-x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases} \qquad 0 < p < 1.$$

    (a) Find $\mathbb{E}(X)$.
    (b) Find the moment generating function, $M_X(t)$. (*Hint:* The mgf will not exist [it will be ∞] for some values of t. Keep that in mind as you solve the problem and be sure to specify the values of t for which $M_X(t)$ does exists.)